

Extra homework 6

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function and let x^* be a solution to the equation $f(x^*) = 0$. We will approximate x^* using a rapidly converging sequence (x_n) , through Newton's method. At each iteration x_n , we consider the linear approximation $T_1(x)$ for $f(x)$ around x_n . We define the next iteration x_{n+1} by imposing the condition $T_1(x_{n+1}) = 0$.

(a) Show that

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

(b) Show that if (x_n) is convergent, then $x_n \rightarrow x^*$.

(c) Define the approximation error $e_n = |x_n - x^*|$. How does e_{n+1} compare to e_n ?

Using Taylor's formula to expand $f(x^*)$ around x_n , show that, for some $c_n \in (x^*, x_n)$,

$$e_{n+1} = \frac{f''(c_n)}{2f'(x_n)} e_n^2.$$

(d) Assume that x^* is a simple root to the equation $f(x^*) = 0$ and that f is twice continuously differentiable in a neighborhood of x^* . If x_0 is sufficiently close to x^* prove that (x_n) converges quadratically to x^* , meaning that there exists $C > 0$ such that

$$e_{n+1} \leq C e_n^2.$$

(e) Using this root finding method, construct a sequence that approximates $\sqrt{a}$, $a > 0$ and illustrate the quadratic convergence in **Python**, showing that the number of correct digits approximately doubles with each iteration.

These questions are extra. You will get bonus points for solving them.